MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW POWER TYPES 2N2432, 2N2432A JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a low power, high speed chopper, NPN, silicon transistor. Four levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (TO-18).

1.3 Maximum ratings.

	P_T 1/ T_A = 25°C	P_T 2/ T_C = 25°C	V_{CBO}	V_{CEO}	V_{ECO}	I_C	T_{STG}	T_J
	mW	mW	Vdc	Vdc	Vdc	mAdc	°C	°C
2N2432	300	600	30	30	15	100	-65 to +200	-65 to +175
2N2432A	300	600	45	45	18	100	-65 to +200	-65 to +175

1/ Derate linearly 2.0 mW/°C for T_A >25°C.

2/ Derate linearly 4.0 mW/°c for T_C >25°C.

1.4 Primary electrical characteristics at T_A = 25°C.

	h_{FE1}	h_{FE2}	$h_{FE(inv)1}$		$V_{CE(sat)}$	$r_{ec(on)}$		$R_{\theta JC}$
	V_{CE} = 5 Vdc I_C = 10 µAdc	V_{CE} = 5 Vdc I_C = 1 mAdc	V_{EC} = 5 Vdc I_E = 200 µAdc		I_C = 10 mAdc I_B = 500 µAdc	I_e = 100 µA ac(rms) I_B = 1 mAdc I_E = 0 f = 1 kHz		
			2N2432	2N2432A		2N2432	2N2432A	
					Vdc	Ohms	Ohms	°C/mW
Min	30	80	2	3	---	---	---	---
Max	---	400	---	---	0.15	20	15	0.25

	$V_{EC(ofs)1}$ I_B = 200 µAdc I_E = 0		$V_{EC(ofs)2}$ I_B = 1.0 mAdc I_E = 0		$\lvert h_{fe} \rvert$ V_{CE} = 5 Vdc I_C = 1 mAdc f = 20 MHz	C_{obo} V_{CB} = 0; I_E = 0 100 kHz ≤ f ≤ 1 mHz	C_{ibo} V_{EB} = 0; I_C = 0 100 kHz ≤ f ≤ 1 MHz
	2N2432	2N2432A	2N2432	2N2432A			
	mVdc	mVdc	mVdc	mVdc		pF	pF
Min	---	---	---	---	2	---	---
Max	0.5	0.4	1.0	0.7	10	12	12

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, ATTN: RBE-2, Griffiss AFB, NY 13441 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications, standards, and handbooks, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500, and as follows:

$V_{(BR)ECO}$ - - - -	Breakdown voltage, emitter to collector, with base open-circuited.
$h_{FE(inv)}$ - - - -	Forward-current transfer ratio except that the collector and emitter shall be interchanged, i.e., I_E/I_B.
I_e - - - - - - - -	Emitter current (rms).
I_{ECS} - - - - - -	Emitter to collector cutoff current, with base short-circuited to collector.
$r_{ec(on)}$ - - - - -	Small-signal emitter-collector on-state resistance.
V_{BC} - - - - - - -	Base to collector voltage.
$V_{EC(ofs)}$ - - - -	Emitter to collector offset voltage, i.e., open circuit voltage between emitter and collector when the base-collector junction is forward biased.
V_{ec} - - - - - - -	Emitter to collector voltage (rms).

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1 herein.

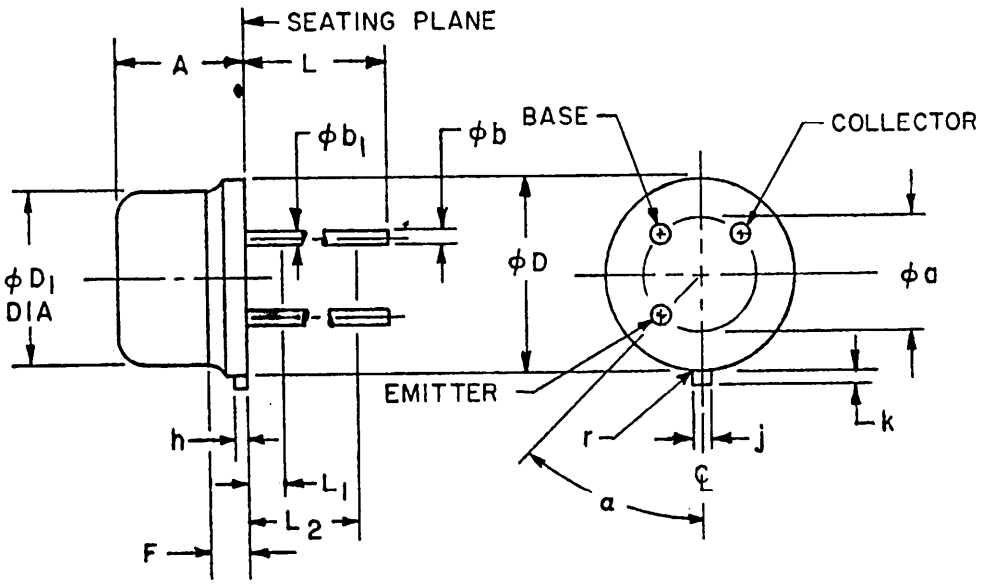
3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be gold or tin plated, or solder dipped. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order.

3.4 Marking. Marking shall be marked in accordance with MIL-S-19500. At the option of the manufacturer, the following marking may be omitted from the body of the transistor:

a. Country of origin.

b. Manufacturer's identification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

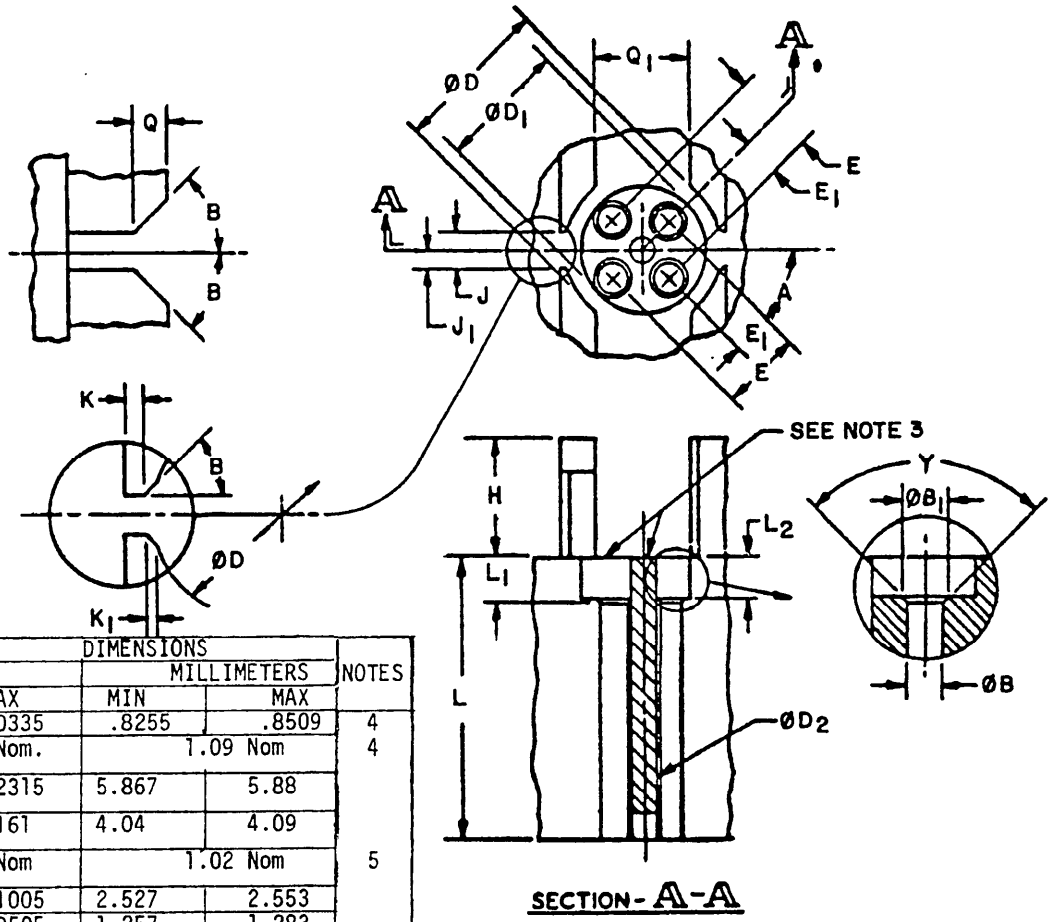


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.170	.210	4.32	5.33	
Øa	.100 TP		2.54 TP		5
Øb	.016	.021	.41	.53	6,7
$Øb_1$	.016	.019	.41	.48	6,7
ØD	.209	.230	5.31	5.84	
$ØD_1$	.178	.195	4.52	4.95	
F	---	.030	---	.76	4
h	---	.020	---	.51	
j	.036	.046	.91	1.17	2
k	.028	.048	.71	1.22	2,3
L	.500	---	12.70	---	6,7
L_1	---	.050	----	1.27	6,7
L_2	.250	---	6.35	---	6,7
r	---	.010	- -	.25	9
α	45° TP		45° TP		5

NOTES:
1. Metric equivalents are given for general information only and are based upon 1 inch = 24.5 mm.
2. Beyond r (radius) maximum, j shall be held for a minimum length of .011 (.28mm).
3. k measured for maximum ØD.
4. Body contour optional within zone defined by ØD, $ØD_1$ and F.
5. Leads at gage plane .054 +.001 -.000 (1.37 +.03 -.00mm) below seating plane shall be within .007 (.18mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gage and gaging procedure shown in figure 2.
6. $Øb_1$ applies between L_1 and L_2. Øb applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond L minimum.
7. All three leads.
8. The collector shall be internally connected to the case.
9. r (radius) applies to both inside corners of tab.

FIGURE 1. Physical dimensions (TO-18).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.



LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
ØB	.0325	.0335	.8255	.8509	4
$ØB_1$	.043 Nom.		1.09 Nom		4
ØD	.2310	.2315	5.867	5.88	
$ØD_1$	.159	.161	4.04	4.09	
$ØD_2$	.040 Nom		1.02 Nom		5
E	.0995	.1005	2.527	2.553	
E_1	.0495	.0505	1.257	1.283	
H	.145	.155	3.68	3.94	
J	.0470	.0475	1.194	1.207	
J_1	.0235	.0245	.597	.622	
K	.009	.011	.229	.279	
K_1	.005 Nom		.127 Nom		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
L_2	.043 Nom		1.09 Nom		
Q	.040 Nom		1.02 Nom		
Q_1	.123	.127	3.12	3.23	
A	44.90°	45.10°	---	---	
B	45° Nom		---	---	
Y	90° Nom		---	---	

NOTES:
1. Metric equivalents are given for general information only and are based upon 1 inch = 24.5 mm.
2. The following gaging procedure shall be used: The device being measured shall be inserted until its seating plane is .125 (3.18 mm) + .010 (.254 mm) from the seating surface of the gage. A force of 8±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plant of the device shall be seated against the gage.
   The use of a pin straightener prior to insertion in the gage is permissible.
   A spacer may be used to obtain the .125 (3.18mm) distance from the gage seat prior to force application.
3. These surfaces to be parallel and in same plane within ±.001 (.025 mm).
4. Four holes.
5. Pressed in.

FIGURE 2. Gage for lead and tab location.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{CBO1} and h_{FE1}	Not applicable
11	I_{CBO1} and h_{FE1}; ΔI_{CBO1} = 100% or 5 nAdc, whichever is greater; Δh_{FE1} = ±15%	I_{CBO1} and h_{FE1}
12	See 4.2.1	See 4.2.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO1} = 100% or 5 nAdc, whichever is greater; Δh_{FE1} = ±15%	Subgroup 2 of table I herein; ΔI_{CBO1} = 100% or 5 nAdc, whichever is greater; Δh_{FE1} = ±25%

4.2.1 Power burn-in conditions. Power burn-in conditions are as follows:

JANS level: V_{CB} = 10 Vdc; P_T = 300 mW at T_A = 25°C.

JANTX and JANTXV levels:

V_{CB} = 15 Vdc (2N2432) P_T = 300 mW at T_A = 25°C

V_{CB} = 25 Vdc (2N2432A)

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Emitter to collector cutoff current. Method of test shall be in accordance with method 3041 of MIL-STD-750, condition C, except that all references to the collector and emitter of the transistor under test shall be interchanged.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

4.5.3 Emitter to collector breakdown voltage. Method of test shall be in accordance with method 3011 of MIL-STD-750, condition D, except that all references to the collector and emitter of the transistor under test shall be interchanged.

4.5.4 Forward current transfer ratio (inverted connection). Method of test shall be in accordance with method 3076 of MIL-STD-750, except that all references to the collector and the emitter of the transistor under test shall be interchanged. Then: $h_{FE(inv)} = I_E/I_B$.

4.5.5 Emitter to collector offset voltage. The transistor shall be tested in the circuit of figure 3. The base current shall be adjusted to the specific value. The voltage between the emitter and collector shall be measured using a voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement.

4.5.6 Small-signal emitter-collector on-state resistance. The transistor shall be tested in the circuit of figure 4. The base current shall be adjusted to the specific value and an ac sinusoidal signal current, I_e, of the specified rms value shall be applied between the emitter and collector. The rms voltage V_{ec}, between the emitter and collector shall be measured using an ac voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement. The small-signal emitter-collector on-state resistance shall then be determined as follows:

$$r_{ec(on)} = \frac{V_{ec}}{I_E}$$

where V_{ec} is the rms voltage between the emitter and collector.

4.5.7 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 30 mAdc.

b. Collector emitter voltage magnitude shall be 10 Vdc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be selected with $25°C \leq T_R \leq 75°C$ and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. Maximum limit for $R_{\theta JC}$ shall be 0.25°C/mW.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Acquisition documents should specify the following:

a. Lead finish (see 3.3.1).

b. Lead material (see 3.3.1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - MSFC - EG02

Review activities:
Army - AR, AV, MI
Navy - SH
Air Force - 11, 85
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 14, 19

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5961-0830)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Conditions	JANS	JAN JANTX JANTXV		Min	Max	
Subgroup 1								
Visual and mechanical inspection	2071		15 (c=0)	5				
Subgroup 2								
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mAdc pulsed (see 4.5.1)	3 1/	5	$V_{(BR)CEO}$			
2N2432						30	---	Vdc
2N2432A						45	---	Vdc
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 100 µAdc			$V_{(BR)CBO}$			
2N2432						30	---	Vdc
2N2432A						45	---	Vdc
Breakdown voltage, emitter to collector	3011	Bias condition D; I_E = 100 µAdc; I_B = 0 (see 4.5.3)			$V_{(BR)ECO1}$			
2N2432						15	---	Vdc
2N2432A						18	---	Vdc
Breakdown voltage, emitter to collector	3011	Bias condition D; I_E = 10 mAdc; I_B = 0; pulsed (see 4.5.1 and 4.5.3)			$V_{(BR)ECO2}$	10	---	Vdc
Collector to base cutoff current	3036	Bias condition D;			I_{CBO}			
2N2432		V_{CB} = 25 Vdc				---	10	nAdc
2N2432A		V_{CB} = 40 Vdc				---	10	nAdc
Collector to emitter cutoff current	3041	Bias condition C;			I_{CES1}			
2N2432		V_{CE} = 25 Vdc				---	10	nAdc
2N2432A		V_{CE} = 40 Vdc				---	10	nAdc
Emitter to collector cutoff current	3041	Bias condition C; V_{EC} = 15 Vdc; V_{BC} = 0 (see 4.5.2)			I_{ECS1}	---	2	nAdc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 15 Vdc			I_{EBO}	---	2	nAdc
Forward current transfer ratio	3076	V_{CE} = 5 Vdc; I_C = 10 µAdc			h_{FE1}	30	---	---
Forward-current transfer ratio	3076	V_{CE} = 5 Vdc; I_C = 1 mAdc			h_{FE2}	80	400	---

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		ETPD		Symbol	Limits		Unit
	Method	Conditions	JANS	JAN JANTX JANTXV		Min	Max	
Subgroup 2 - Continued								
Forward-current transfer ratio (inverted connection)	3076	V_{EC} = 5 Vdc; I_E = 0.2 mAdc (see 4.5.4)			$h_{FE(inv)1}$			
2N2432						2	---	---
2N2432A						3	---	---
Saturation voltage and resistance	3071	I_C = 10 mAdc; I_B = 0.5 mAdc			$V_{CE(sat)}$	---	0.15	Vdc
Emitter to collector offset voltage	---	I_B = 200 µAdc; I_E = 0 (see 4.5.5 and figure 3)			$V_{EC(ofs)1}$			
2N2432						---	0.5	mVdc
2N2432A						---	0.4	mVdc
Emitter to collector offset voltage	---	I_B = 1 mAdc; I_E = 0 (see 4.5.5 and figure 3)			$V_{EC(ofs)2}$			
2N2432						---	1.0	mVdc
2N2432A						---	0.7	mVdc
Subgroup 3				5				
High temperature operation:		T_A = 125°C						
Collector to emitter cutoff current	3041	Bias condition C;			I_{CES2}			
2N2432		V_{CE} = 25 Vdc				---	250	nAdc
2N2432A		V_{CE} = 40 Vdc				---	250	nAdc
Emitter to collector cutoff current	3041	Bias condition C; V_{EC} = 15 Vdc; V_{BC} = 0 (see 4.5.2)			I_{ECS2}	---	200	nAdc
Low temperature operation:		T_A = -55°C						
Forward-current transfer ratio	3076	V_{CE} = 5 Vdc; I_C = 10 µAdc			h_{FE3}	10	---	---
Forward-current transfer ratio	3076	V_{CE} = 5 Vdc; I_C = 1 mAdc			h_{FE4}	25	---	---
Forward-current transfer ratio (inverted connection)	3076	V_{EC} = 5 Vdc; I_E = 200 µAdc (see 4.4.4)			$h_{FE(inv)2}$	1.8	---	---

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Conditions	JANS	JAN JANTX JANTXV		Min	Max	
Subgroup 4								
Small signal emitter collector on state resistance	---	I_B = 1 mAdc; I_E = 0; I_e = 100 µAac(rms); f = 1 kHz (see 4.5.6 and figure 4)			$r_{ec(on)}$			
2N2432						---	20	Ohms
2N2432A						---	15	Ohms
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 5 Vdc; I_C = 1 mAdc; f = 20 mHz			$\|h_{fe}\|$	2	10	---
Open circuit output capacitance	3236	V_{CB} = 0; I_E = 0; 100 kHz $\leq$ f $\leq$ 1 MHz			C_{obo}	---	12	pF
Input capacitance	3240	V_{EB} = 0; I_C = 0; 100 kHz $\leq$ f $\leq$ 1 MHz			C_{ibo}	---	12	pF
Subgroups 5, 6, 7								
Not applicable								

1/ For JANS level, all devices required by the specific LTPD shall be subjected to subgroups 2, 3, and 4 combined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

TABLE IIa. Group B inspection for JANS devices.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD	Small lot quality conformance inspection n/c
	Method	Conditions		
Subgroup 1			10	8/0
Physical dimensions	2066			
Subgroup 2			15	6/0
Solderability	2026			
Resistance to solvents	1022			
Subgroup 3			10	6/0
Thermal shock (temperature cycling)	1051			
Hermetic seal	1071			
a. Fine leak b. Gross leak				
Electrical measurements		See table IV, steps 1, 3, 4, 6, 8, 9, and 10.		
Decap internal visual design verification	2075			
Bond strength	2037	Test condition A; all internal wires for each device shall be pulled separately.		
Die shear strength	2017	Only devices previously subjected to bond strength test shall be used for this test.		
Subgroup 4			10	N/A
Intermittent operation life	1037	V_{CB} = 10 Vdc; P_T = 300 mW at T_A = 25°C; $t_{on} = t_{off}$ = 3 minutes minimum for 2000 cycles. No heat sink or forced air cooling on the devices shall be permitted.		
Electrical measurements		See table IV, steps 1, 3, 4, 6, 8, 9, 10, and 14.		
Subgroup 5			10	12/2
Accelerated steady-state operation life	1027	V_{CB} = 10 Vdc; T_A = 125 ± 25°C for 96 hours; P_T = 300 mW at T_A = 100°C or adjusted as required by the chosen T_A to give an average lot T_J = 275°C. Marking legibility requirements shall not apply.		

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

TABLE IIa. Group B inspection for JANS devices - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD	Small lot quality conformance inspection n/c
	Method	Conditions		
Subgroup 5 - Continued			10	8/0
Electrical measurements		See table IV, steps 1, 3, 4, 6, 8, 9, 10, 12, and 13.		
Bond strength (Al-Au die interconnects only)	2037	Test condition A	20 (c = 0)	5/0
Subgroup 6			10	8/0
Thermal resistance	3131	See 4.5.7		

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

FIGURE 2. Thermal

TABLE IIb. Group B inspection for JAN, JANTX, and JANTXV devices.

Inspection	MIL-STD-750		LTPD
	Method	Conditions	
Subgroup 1			15
Solderability	2026		
Resistance to solvents	1022		
Subgroup 2			10
Thermal shock (temperature cycling)	1051		
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table IV, steps 1, 4, 6, and 10.	
Subgroup 3			5
Steady-state operation life	1027	P_T = 300 mW; V_{CB} = 15 Vdc (2N2432); V_{CB} = 25 Vdc (2N2432A)	
Electrical measurements		See table IV, steps 2, 5, 7, and 11.	
Subgroup 4			
Decap internal visual design verification	2075		1 device/0 failure for each lot.
Bond strength	2037	Test condition A; all internal wires for each device shall be pulled separately.	20 (c = 0)
Subgroup 5			15
Thermal resistance	3131	See 4.5.7	
Subgroup 6			
High-temperature life (non-operating)	1032	T_{STG} = 200°C	7
Electrical measurements		See table IV, steps 2, 5, 7, and 11.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

TABLE III. Group C inspection (all quality levels).

Inspection	MIL-STD-750		LTPD
	Method	Conditions	
Subgroup 1			15
Physical dimensions	2066	See figure 1	
Subgroup 2			10
Thermal shock (glass strain)	1056		
Terminal strength	2036	Test condition E	
Hermetic seal	1071		
Fine leak Gross leak			
Moisture resistance	1021		
External visual	2071		
Electrical measurements		See table IV, steps 1, 3, 4, 6, 8, 9, and 10 (JANS), and steps 1, 4, 6, and 10 (JAN, JANTX, and JANTXV).	
Subgroup 3			10
Shock	2016		
Vibration, variable frequency	2056		
Constant acceleration	2006		
Electrical measurements		See table IV, steps 1, 3, 4, 6, 8, 9, and 10 (JANS), and steps 1, 4, 6, and 10 (JAN, JANTX, and JANTXV).	
Subgroup 4			15
Salt atmosphere (corrosion)	1041		
Subgroup 5			
Not applicable			
Subgroup 6			$\lambda = 10$
Steady-state operation life	1026	P_T = 300 mW; V_{CB} = 15 Vdc (2N2432); V_{CB} = 25 Vdc (2N2432A)	
Electrical measurements		See table IV, steps 1, 3, 4, 6, 8, 9, 10, 12, and 13 (JANS) and steps 2, 5, 7, and 11 (JAN, JANTX, and JANTXV).,	

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

TABLE IV. Groups B and C electrical measurements.

Step	Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
1.	Collector to base cutoff current	3036	Bias condition D	I_{CBO}			
	2N2432		V_{CB} = 25 Vdc		---	10	nAdc
	2N2432A		V_{CB} = 40 Vdc		---	10	nAdc
2.	Collector to base cutoff current	3036	Bias condition D	I_{CBO}			
	2N2432		V_{CB} = 25 Vdc		---	20	nAdc
	2N2432A		V_{CB} = 40 Vdc		---	20	nAdc
3.	Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 15 Vdc	I_{EBO}	---	2	nAdc
4.	Emitter to collector cutoff current	3041	Bias condition C; V_{EC} = 15 Vdc; V_{BC} = 0 (see 4.5.2)	I_{ECS1}	---	2	nAdc
5.	Emitter to collector cutoff current	3041	Bias condition C; V_{EC} = 15 Vdc; V_{BC} = 0 (see 4.5.2)	I_{ECS1}	---	4	nAdc
6.	Forward-current transfer ratio	3076	V_{CE} = 5 Vdc; I_C = 10 µAdc	h_{FE1}	30	---	---
7.	Forward-current transfer ratio	3076	V_{CE} = 5 Vdc; I_C = 10 µAdc	h_{FE1}	24	---	---
8.	Forward-current transfer ratio	3076	V_{CE} = 5 Vdc; I_C = 1 mAdc	h_{FE2}	80	400	---
9.	Forward-current transfer ratio (inverted connection)	3076	V_{EC} = 5 Vdc; I_E = 0.2 mAdc (see 4.5.4)	$h_{FE(inv)1}$			
	2N2432				2	---	---
	2N2432A				3	---	---
10.	Emitter to collector offset voltage	---	I_B = 200 µAdc; I_E = 0; (see 4.5.5 and figure 3)	$V_{EC(ofs)1}$			
	2N2432				---	0.5	mVdc
	2N2432A				---	0.4	mVdc
11.	Emitter to collector offset voltage	---	I_B = 200 µAdc; I_E = 0; (see 4.5.5 and figure 3)	$V_{EC(ofs)1}$			
	2N2432				---	0.6	mVdc
	2N2432A				---	0.7	mVdc

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

TABLE IV. Groups B and C electrical measurements - Continued.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
12.	Collector to base cutoff current	3036	Bias condition D	ΔI_{CBO} 1/		100% or 8 nAdc, whichever is greater	
13.	Forward current transfer ratio	3076	V_{CE} = 5 Vdc; I_C = 10 µAdc	Δh_{FE1} 1/		±25% from initial reading	
14.	Saturation voltage and resistance	3071	I_C = 10 mAdc; I_B = 0.5 mAdc	$\Delta V_{CE(sat)}$ 1/	---	±50	mVdc

1/ Devices that exceed that group A limits for this test shall not be acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

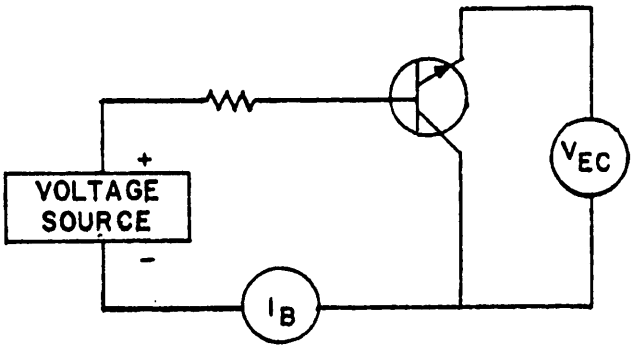


FIGURE 3. Emitter to collector offset voltage test circuit.

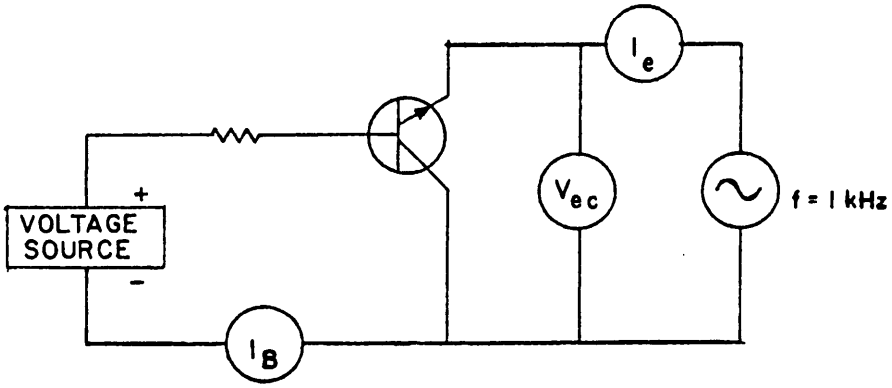


FIGURE 4. Small-signal emitter-collector on-state resistance test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-19500/313B, Semiconductor Device, Transistor, NPN, Silicon, Low Power Types 2N2432, 2N2432AJAN, TX, TXV, AND JANS

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

Commander
Rome Air Development Center
Attn: RBE-2
Griffiss AFB, NY 13441

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:16Z
Check the source to verify that this is the current version before use.